The Mordell-Weil Theorem

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Abstract

The goal of this lecture is to prove the full Mordell-Weil Theorem which states the group of rational points on an elliptic curve over a number field is finitely generated. We do this by way of *descent*, leveraging the fact from the weak Mordell-Weil Theorem that many of the quotients of the group of rational points are finite. However, in order to apply the method of descent, we must first develop the theory of *heights* on an elliptic curve.

1 Descent and the Proof of the Mordell-Weil Theorem

We begin by recalling the statement of the Mordell-Weil Theorem and the weak Mordell-Weil Theorem:

Theorem 1 (Mordell-Weil)**.** *Let K be a number field, and E/K an elliptic curve. Then the group $E(K)$ is finitely-generated.*

Theorem 2 (Weak Mordell-Weil)**.** *Let K be a number field, let E/K be an elliptic curve, and let $m \geq 2$ be an integer. Then $E(K)/mE(K)$ is finite.*

This is the result that we proved last week, and the issue we ran into to is that such a result tells us nothing about whether the group is finitely-generated or not. The crux of the matter is there may be too much m-divisibility in the group.

The method of descent tells us that under certain circumstances (like having a suitable *height* function), m-divisibility isn't an issue.

Let me quickly describe the idea of descent: Take a group A so that A/mA is finite. We can picture A/mA as lying on an x-axis, and then A lying in the upper half plane. The height of a point $P \in A$ can be seen then as the y-value of the point. Now, in the case of a metric, we would expect the height of a point P to be less than the height of mP.

So starting at P, project down to the x-axis to get $Q_1 \in A/mA$. Then we can write $P = mP_1 + Q_1$, and we would expect the height of mP_1 to be similar to that of P. Then the height of P_1 is much less. Continue this process until you get arbitrarily close to the x-axis. If the set of points in A sitting close to the x-axis is finite, then by taking the Q_i and the points close to the x-axis, we see A is finitely generated.

This is the process of descent, and we can make the statement rigorous:

Theorem 3 (Descent)**.** *Let A be an abelian group. Suppose there is a* height *function*

$$h : A \to \mathbb{R}$$

satisfying:

1. *For $Q \in A$, there is a constant C_1 depending only on Q such that*

$$h(P+Q) \leq 2h(P) + C_1$$

2. *There is an integer $m \geq 2$ and constant C_2 independent of P for which*

$$h(mP) \geq m^2 h(P) - C_2$$

3. *For every constant C_3, the set*

$$\{P \in A : h(P) \leq C_3\}$$

is finite.

Then if for the same m in (ii) the group A/mA is finite, then A is finitely generated.

Note then that the full Mordell-Weil Theorem may be deduced from the weak form so long as we have a suitable height function to use descent on. Developing this height function is the goal for the rest of this note.

2 Heights on Elliptic Curves

Before restricting to elliptic curves, we first define heights on general projective space.

Definition 1. Let $P \in \mathbb{P}^N(K)$ be a point in projective space with a fixed set of homogeneous coordinates.

Writing $P = [x_0 : \cdots : x_N]$ with $x_i \in K$, we define the *height of P* to be

$$H_K(P) = \prod_{\nu \in M_K} \max\left\{|x_0|_\nu, \ldots, |x_N|_\nu\right\}^{n_\nu}$$

where n_ν is the local degree.

Theorem 4 (VIII 5.4)**.** *Take $P \in \mathbb{P}^N(K)$.*

1. *The height is independent of choice of homogeneous coordinates.*
2. *$H_K(P) \geq 1$.*
3. *If L/K is a finite extension, $H_L(P) = H_K(P)^{[L:K]}$*

Proof. 1. If we multiply each coordinate by a non-zero constant λ, this pulls out to be the same as multiplying the height by

$$\prod_{\nu \in M_K} \lambda^{n_\nu}$$

2. Without loss of generality make one of the coordinates 1.

3. We may break the product into

$$\prod_{\mu \in M_L} \max\left\{|x_0|_\mu, \ldots, |x_N|_\mu\right\}^{n_\mu} = \prod_{\nu \in M_K} \prod_{\mu|\nu} \max\left\{|x_0|_\mu, \cdots, |x_N|_\mu\right\}^{n_\mu}$$

And each

$$\prod_{\mu|\nu} \max\left\{|x_0|_\mu, \ldots, |x_N|_\mu\right\}^{n_\mu} = \prod_{\mu|\nu} \max\left\{|x_0|_\mu, \ldots, |x_N|_\mu\right\}^{n_\mu} = \max\left\{|x_0|_\nu, \ldots, |x_N|_\nu\right\}^{[L:K]n_\nu}$$

And from this the exponent pulls out.

□

The limitation that we have with this way of defining height is that it depends on the field of definition of a point. What we are looking for is some *geometric* definition that defines the height for any point on $E(\overline{K})$. The last part of the previous proposition suggests we should define it by taking a root with respect to the degree of the field of definition over $\mathbb{Q}$.

Definition 2. The *absolute height* of a point $P \in E(\overline{K})$ is

$$H(P) := H_L(P)^{\frac{1}{[L:\mathbb{Q}]}}$$

for any field L on which P is defined.

A natural question to ask is how the height of a point changes under a morphism of degree d. The answer is that the absolute height acts like a degree d polynomial.

Theorem 5 (VIII 5.6)**.** *Let $F : \mathbb{P}^N \to \mathbb{P}^M$ be a morphism of degree d. Then there are positive constants C_1 and C_2 depending only on F such that*

$$C_1 H(P)^d \leq H(F(P)) \leq C_2 H(P)^d$$

The next property regarding heights we want to explore is how the height of the roots of a polynomial compare to the heights of the coefficients.

Theorem 6 (VIII 5.9)**.** *Let*

$$f(T) = \sum_{i=0}^{d} a_i T^{d-i} = a_0(T - \alpha_1) \cdots (T - \alpha_d)$$

be a polynomial of degree d. Then

$$2^{-d} \prod_{j=1}^{d} H(\alpha_j) \leq H([a_0, \ldots, a_d]) \leq 2^{d-1} \prod_{j=1}^{d} H(\alpha_j)$$

Finally, we prove a key finiteness result of the heights of points in projective space.

Theorem 7 (VIII 5.11)**.** *Let C and d be constants. Then the set*

$$\left\{P \in \mathbb{P}^N(K) : H_K(P) \leq C\right\}$$

is a finite set.

3 Proof of the Mordell-Weil Theorem

As we saw in the first section, we just need to prove the existence of a height function on an elliptic curve which satisfies the conditions of the Descent Theorem for some m.

The height function we take is the following:

Definition 3. Let H be the height on projective space. This is mostly multiplicative, so we change it to an additive function by defining

$$h(P) = \log H(P)$$

This is called the *absolute logarithmic height*.

For E/K an elliptic curve, and $f \in \bar{K}(E)$, The *height on E relative to f* is

$$h_f : E(\bar{K}) \to \mathbb{R}, \quad h_f(P) = h(f(P))$$

where we view f as a map into $\mathbb{P}^1$.

We now prove the three hypotheses of the Descent Theorem for when $f = x$ (though the results hold for *any* even function f).

Theorem 8. *Let E/K be an elliptic curve, and let $f \in K(E)$ be a non-constant function. Then for any constant C, the set*

$$\{P \in E(K) : h_f(P) \leq C\}$$

is finite.

Proof. As f is defined over K, it maps points of K to points in $\mathbb{P}^1(K)$. This is a surjective map, so f is finite-to-one, and the image of the set in question is

$$\left\{Q \in \mathbb{P}^1(K) : H(Q) \leq e^C\right\}$$

□

Theorem 9. *Let E/K be an elliptic curve, and let $f \in K(E)$ be an even function. Then for all $P, Q \in E(\bar{K})$ we have*

$$h_f(P+Q) + h_f(P-Q) = 2h_f(P) + 2h_f(Q) + O(1)$$

where the constant depends only on E and f. For our needs we only need to prove this for $f = x$.

Proof. As we are in the characteristic 0 case, we can give E/K as cut out by a Weierstrass equation

$$y^2 = x^3 + Ax + B$$

Then $h_x(O) = 0$ and $h_x(P) = h_x(-P)$. (This last equality follows because the explicit inversion formula on the elliptic curve preserves the x-value of any point [not at infinity].) Therefore we see the equality holds with no constant whenever P or Q is the point at infinity. Therefore we may assume $P \neq O \neq Q$.

The heart of this result is determining how $h_x(P) + h_x(Q)$ relate to $h_x(P+Q)$ and $h_x(P-Q)$. This leads us to our main calculation that will motivate the rest of our approach to this theorem.

Note that in the context of this logarithmic absolute height, considering

$$f(T) = (T - x(P))(T - x(Q))$$

we have

$$h_x(P) + h_x(Q) + C_1 \leq h([1, x(P) + x(Q), x(P)x(Q)]) \leq h_x(P) + h_x(Q) + C_2$$

This says that

$$h_x(P) + h_x(Q) = h([1, x(P) + x(Q), x(P)x(Q)]) + O(1)$$

and this holds for any points P, Q not at infinity. Therefore instead of comparing $h_x(P)$, $h_x(Q)$, $h_x(P-Q)$, $h_x(P+Q)$ we may instead compare $h([1, x(P) + x(Q), x(P)x(Q)])$ to $h([1, x(P+Q) + x(P-Q), x(P+Q)x(P-Q)])$.

By playing around with the addition formulas we have the equations

$$x(P+Q) + x(P-Q) = \frac{2(x(P) + x(Q))(A + x(P)x(Q)) + 4B}{(x(P) + x(Q))^2 - 4x(P)x(Q)}$$

and

$$x(P+Q)x(P-Q) = \frac{(x(P)x(Q) - A)^2 - 4B(x(P) + x(Q))}{(x(P) + x(Q))^2 - 4x(P)x(Q)}$$

This means that $[1, x(P+Q) + x(P-Q), x(P+Q)x(P-Q)]$ can be viewed as the image of the morphism

$$g([t, u, v]) = [u^2 - 4tv, 2u(At + v) + 4Bt^2, (v - At)^2 - 4Btu]$$

when $t = 1$, $u = x(P) + x(Q)$ and $v = x(P)x(Q)$. We will show later that g is actually a morphism $\mathbb{P}^2 \to \mathbb{P}^2$.

Assuming that this is the case, as g is a degree 2 map, we have

$$\begin{aligned} h_x(P+Q) + h_x(P-Q) + O(1) &= h([1, x(P+Q) + x(P-Q), x(P+Q)x(P-Q)]) \\ &= 2h([1, x(P) + x(Q), x(P)x(Q)]) + O(1) \\ &= 2h_x(P) + 2h_x(Q) \end{aligned}$$

and this (modulo some details) completes the proof.

We now make this completely rigorous, recognizing that $x(P)$ is not a number but a point in projective space.

With g defined as above, we have a commutative diagram

$$\begin{tikzcd} E \times E \arrow[r, "G"] \arrow[d] \arrow[dd, bend right=60, "\sigma"'] & E \times E \arrow[d] \arrow[dd, bend left=60, "\sigma"] \\ \mathbb{P}^1 \times \mathbb{P}^1 \arrow[d] & \mathbb{P}^1 \times \mathbb{P}^1 \arrow[d] \\ \mathbb{P}^2 \arrow[r, "g"] & \mathbb{P}^2 \end{tikzcd}$$

Where $G(P, Q) = (P + Q, P - Q)$, and σ is given by composing

$$E \times E \to \mathbb{P}^1 \times \mathbb{P}^1, \quad (P, Q) \mapsto (x(P), y(P))$$

with

$$\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^2, \quad ([\alpha_1, \beta_1], [\alpha_2, \beta_2]) \mapsto [\beta_1\beta_2, \alpha_1\beta_2 + \alpha_2\beta_1, \alpha_1\alpha_2]$$

We now want to show that g is a morphism. All that we need to do for this is show the coordinates in the image are never simultaneously 0. First consider the complement of the affine $t = 1$.

Then $g([t,u,v]) = 0$ implies $u = 0$ and $v = 0$. Thus it remains to check on the affine chart $t = 1$. In this case,

$$g(u,v) = [u^2 - 4v, 2u(A+v) + 4B, (v-A)^2 - 4Bu]$$

Substituting $v = \frac{u^2}{4}$, we have $g(u,v) = 0$ if and only if

$$2uA + \frac{u^3}{2} + 4B = 0, \text{ and } \frac{u^4}{16} - \frac{Au^2}{2} + A^2 - 4Bu = 0$$

Replacing $x = \frac{u}{2}$ this is equivalent to

$$4x^3 + 4xA + 4B = 0, \text{ and } x^4 - 2Ax^2 - 8Bx + A^2 = 0$$

Then, noticing that

$$(12x^2+16A)(x^4-2Ax^2-8Bx+A^2)-(3x^3-5Ax-27B)(4x^2+4Ax+4B) = 4(4A^3+27B^2) = 4\Delta \neq 0$$

we conclude that the image of g is always a well-defined point in projective space and is therefore a morphism.

Now g is a morphism of degree 2, so using Theorem VIII.5.6 we have

$$h(\sigma(P+Q, P-Q)) = h(g \circ \sigma(P,Q)) = 2h(\sigma(P,Q)) + O(1)$$

Now we will show that

$$h(\sigma(R_1, R_2)) = h_x(R_1) + h_x(R_2) + O(1)$$

Indeed, this is true if one of $R_i = O$. If this is not the case, we may write $x(R_1) = [\alpha_1, 1]$ and $x(R_2) = [\alpha_2, 1]$. Then

$$h(\sigma(R_1, R_2)) = h([1, \alpha_1 + \alpha_2, \alpha_1\alpha_2])$$

and

$$h_x(R_1) + h_x(R_2) = h(\alpha_1) + h(\alpha_2)$$

Using Theorem VIII.5.9 to the polynomial $(T + \alpha_1)(T + \alpha_2)$ we find

$$h(\alpha_1) + h(\alpha_2) - \log 4 \leq h([1, \alpha_1 + \alpha_2, \alpha_1\alpha_2]) \leq h(\alpha_1) + h(\alpha_2) + \log 2$$

Finally, we use this formula on both sides of the equation

$$h(\sigma(P+Q, P-Q)) = 2h(\sigma(P,Q)) + O(1)$$

to get our result. □

This result then lets us conclude our height function on elliptic curves satisfies the conditions for a height function to be used in the descent procedure.

Corollary 1. *Let E/K be an elliptic curve.*

1. *Let $Q \in E(\overline{K})$. Then*

$$h_x(P+Q) \leq 2h_x(P) + O(1)$$

with the constant only depending on E and Q.

2. *Let $m \in \mathbb{Z}$. Then*
$$h_x([m]Pj) = m^2 h_f(P) + O(1)$$
with the constant depending only on E and m.

Proof. 1. This follows immediately using the fact that $h_x(P - Q) \geq 0$.

2. This follows via induction: As x is even it suffices to show this for only positive m. The case $m = 0, 1$ is trivial. We now suppose this result holds for $m - 1$ and m. Then replacing P, Q in the previous theorem with $[m]P$ and P we have:
$$\begin{aligned} h_x([m+1]P) &= -h_x([m-1]P) + 2h_x([m]P) + 2h_x(P) + O(1) \\ &= (-(m-1)^2 + 2m^2 + 2)h_x(P) + O(1) \\ &= (m+1)^2 h_x(P) + O(1) \end{aligned}$$
as desired

□

We are now able to conclude the proof of the Mordel-Weil Theorem

Theorem 10 (Mordell-Weil)**.** *For K a number field and E/K an elliptic curve, $E(K)$ is finitely-generated.*

Proof. The above shows for $m = 2$, the height function h_x satisfies all the conditions for descent, and by the Weak Mordell-Weil Theorem, $E(K)/2E(K)$ is finite, $E(K)$ is finitely-generated. □